

Linear Regression

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Outline

- 1 Data as Vectors and Matrices
- 2 Simple Linear Regression

Data as Vectors and Matrices

Observations, features and labels

i	features			label
	TV	radio	newspaper	sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
200	232.1	8.6	8.7	13.4

- Each row is one **observation** (a market): $n = 200$ observations
- The columns TV, radio, newspaper are the **features** (predictors, inputs, independent variables): $p = 3$, budgets in thousands of dollars
- The column sales is the **label** (response, output, target, dependent variable): what we want to predict, in thousands of units

Vectors and matrices

- Observation i is a **feature vector** $\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_p^{(i)})^\top \in \mathbb{R}^p$ with label $y^{(i)} \in \mathbb{R}$; for example $\mathbf{x}^{(1)} = (230.1, 37.8, 69.2)^\top$ and $y^{(1)} = 22.1$
- Stacking the n observations gives the **feature matrix** $\mathbf{X} \in \mathbb{R}^{n \times p}$, whose row i is $(\mathbf{x}^{(i)})^\top$, and the **label vector** $\mathbf{y} \in \mathbb{R}^n$

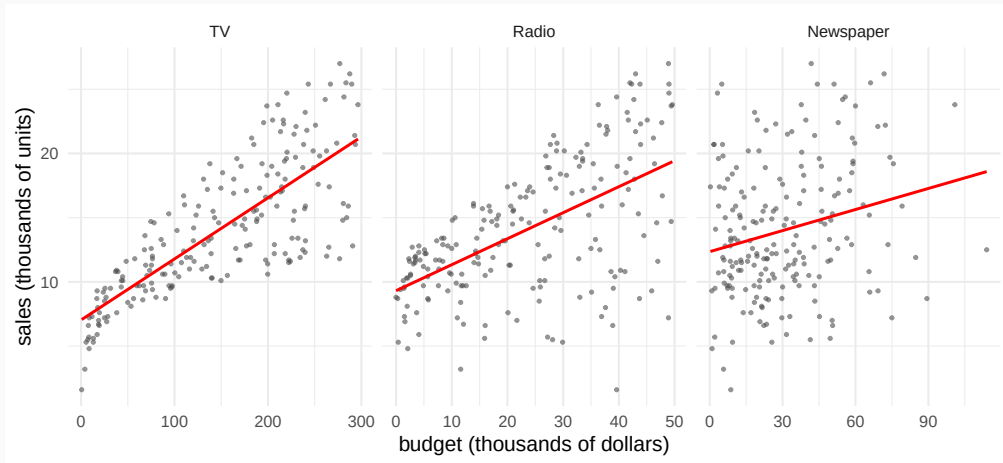
$$\mathbf{X} = \begin{bmatrix} 230.1 & 37.8 & 69.2 \\ 44.5 & 39.3 & 45.1 \\ \vdots & \vdots & \vdots \\ 232.1 & 8.6 & 8.7 \end{bmatrix} = \begin{bmatrix} (\mathbf{x}^{(1)})^\top \\ (\mathbf{x}^{(2)})^\top \\ \vdots \\ (\mathbf{x}^{(n)})^\top \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 22.1 \\ 10.4 \\ \vdots \\ 13.4 \end{bmatrix}$$

- Superscript (i) = which observation (row), subscript j = which feature (column): $x_j^{(i)}$ is entry (i, j) of $\mathbf{X}$; bold lowercase = column vector, bold uppercase = matrix, plain letter = single number

The regression problem

- Assume the label depends on the features through $y = f(\mathbf{x}) + \varepsilon$: f is a fixed but unknown function and ε is a random error with mean zero, independent of $\mathbf{x}$
- **Supervised learning**: estimate f from the labelled data $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$, which gives an estimate $\hat{f}$
- **Regression** when the label is quantitative (sales, mpg, balance); classification when it is qualitative (later lectures)
- **Prediction**: compute $\hat{y} = \hat{f}(\mathbf{x})$ for new observations, treating $\hat{f}$ as a black box
- **Inference**: understand how the label changes with each feature (which features matter, how large the effects are, whether the relationship is linear)

Advertising data: questions regression can answer



Advertising data: questions regression can answer (cont.)

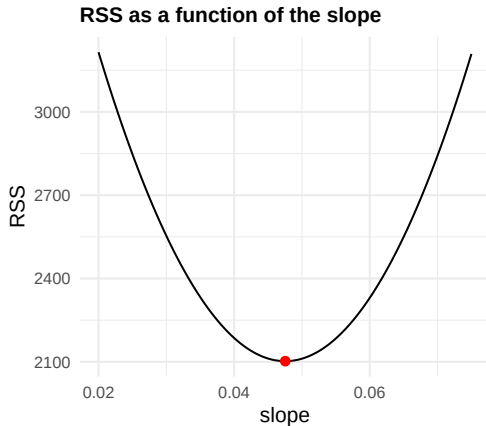
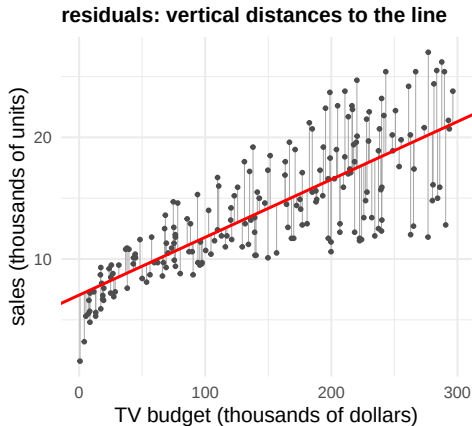
- Is there a relationship between the advertising budgets and sales, and how strong is it?
- Which media are associated with sales, and by how much does each contribute?
- How accurately can we predict sales for given budgets?
- Is the relationship linear, and do the media reinforce each other (synergy)?

Simple Linear Regression

Simple linear regression model

- One feature x and a quantitative label y , related by a straight line plus error: $y = \beta_0 + \beta_1 x + \varepsilon$
- β_0 is the **intercept** (the value of y when $x = 0$), β_1 is the **slope** (the change in y for a one-unit increase in x); together they are the **coefficients**
- Estimate the coefficients from the training data (hat = estimated): $\hat{\beta}_0, \hat{\beta}_1$; predict with $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$
- **Fitted value** of observation i : $\hat{y}^{(i)} = \hat{\beta}_0 + \hat{\beta}_1 x^{(i)}$; **residual**: $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$; residuals are computed from data, the errors $\varepsilon^{(i)}$ are never observed
- Example: $\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}$

Least squares criterion



Least squares criterion (cont.)

- **Residual sum of squares:**

$$\text{RSS} = \sum_{i=1}^n (e^{(i)})^2 = \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x^{(i)})^2$$

- **Least squares** chooses $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS: the line with the smallest total squared vertical distance to the points
- Squaring stops positive and negative residuals from cancelling and penalizes large ones more; the red dot marks the minimum (RSS = 2102.5 for TV)

Least squares estimates

Setting the partial derivatives of RSS with respect to β_0 and β_1 to zero gives closed-form estimates, where $\bar{x}$ and $\bar{y}$ are the sample means:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

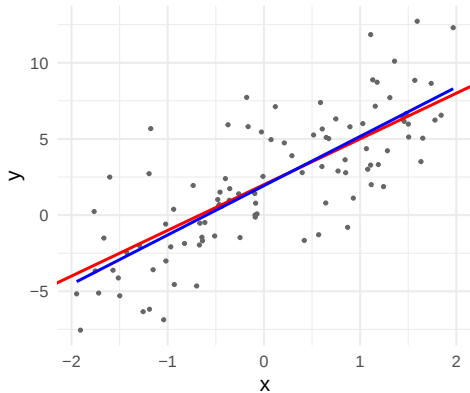
- $\hat{\beta}_1$ is the sample covariance of x and y divided by the sample variance of x
- The fitted line always passes through the point of means $(\bar{x}, \bar{y})$
- Advertising, sales on TV: $\hat{\beta}_0 = 7.03$ and $\hat{\beta}_1 = 0.0475$

Interpreting the coefficients

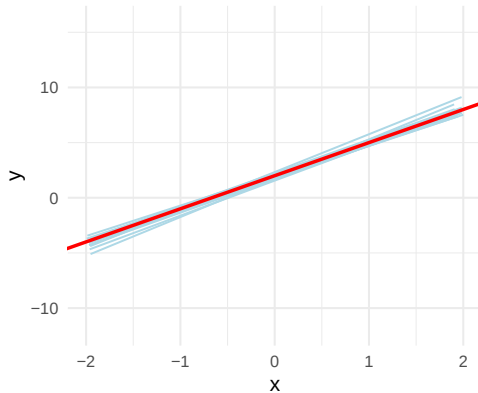
- Fitted line for the Advertising data: $\widehat{\text{sales}} = 7.03 + 0.0475 \times \text{TV}$ (sales in thousands of units, TV budget in thousands of dollars)
- $\hat{\beta}_1 = 0.0475$: an additional \$1,000 of TV budget is associated with about $0.0475 \times 1000 = 47.5$ additional units sold
- $\hat{\beta}_0 = 7.03$: about 7,030 units are predicted when the TV budget is zero; the smallest TV budget in the data is \$700, so the intercept sits at the edge of the data
- The fit describes association in the observed markets; a causal reading needs more than a fitted line, because budgets were not assigned at random

How accurate are the estimates?

one sample: true line (red), fitted line (blue)



ten samples: ten different fitted lines



How accurate are the estimates? (cont.)

- The population regression line (red) is the true, unknown relationship; the least squares line (blue) is computed from one sample, and every new sample gives a different line
- On average the estimates are correct (unbiased), but a single $\hat{\beta}_1$ differs from β_1 ; the **standard error** measures the typical size of this difference

Standard errors of the coefficients

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}, \quad \text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2} \right]$$

- Here $\sigma^2 = \text{Var}(\varepsilon)$ is the variance of the errors; standard errors shrink with more data, less noise, and more spread-out $x^{(i)}$
- σ^2 is unknown and is estimated from the residuals by $\text{RSE}^2 = \text{RSS}/(n-2)$; replacing σ by RSE gives the estimated standard errors reported by software
- These formulas assume the errors are uncorrelated with constant variance σ^2 (checked with residual plots later)
- Advertising, sales on TV: $\text{SE}(\hat{\beta}_0) = 0.4578$ and $\text{SE}(\hat{\beta}_1) = 0.0027$

Confidence intervals and hypothesis tests (recap)

- **95% confidence interval** for β_1 : approximately $\hat{\beta}_1 \pm 2 \cdot \text{SE}(\hat{\beta}_1)$ (exactly, the 0.975 quantile of the t_{n-2} distribution replaces 2)
- **Hypothesis test**: $H_0: \beta_1 = 0$ (no relationship) against $H_a: \beta_1 \neq 0$, using $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$, which follows a t_{n-2} distribution if H_0 is true
- The **p -value** is the probability of a $|t|$ at least this large if H_0 were true; a small p -value (say below 0.05) is evidence against H_0
- Advertising, sales on TV: 95% confidence interval for β_1 is [0.0422, 0.0528]

	coefficient	std. error	t -statistic	p -value
intercept	7.0326	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Assessing the fit: RSE and R-squared

- **Residual standard error:** $RSE = \sqrt{RSS/(n-2)}$, the typical size of a residual, in the units of y ; Advertising: $RSE = 3.26$, about 3,260 units (23% of the mean sales)
- R^2 : the proportion of the variation in y explained by the regression

$$R^2 = 1 - \frac{RSS}{TSS}, \quad TSS = \sum_{i=1}^n (y^{(i)} - \bar{y})^2$$

- $0 \leq R^2 \leq 1$; Advertising: $R^2 = 1 - 2102.5/5417.2 = 0.612$, so TV explains about 61% of the variance in sales
- In simple regression $R^2 = r^2$, the squared correlation between x and y ($r = 0.782$)
- RSE is in the units of y , R^2 is unit-free; a good R^2 depends on the application

Questions?
